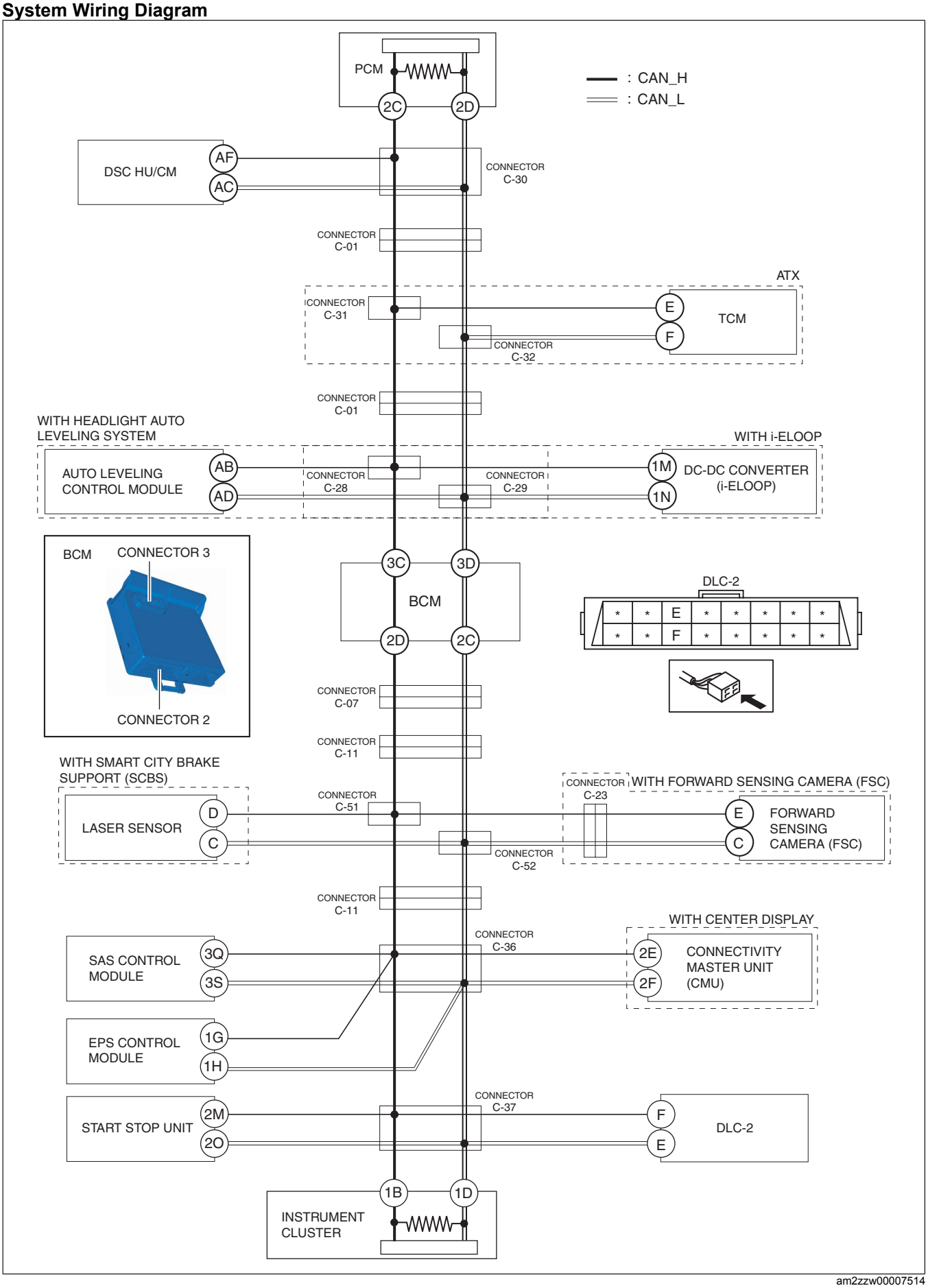
DETERMINING SHORT BETWEEN CIRCUITS LOCATION (HS-CAN) [SKYACTIV-D 1.5]

id100218000800

Caution

- **Perform the following malfunction diagnosis only when it is diagnosed with a short between circuits by CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW. (See CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW [SKYACTIV-D 1.5].)**

System Wiring Diagram



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Determination Procedure

Caution

- **When disconnecting the connector, verify that there is no looseness, damage, deformation, corrosion, or poor connection of the connector terminals.**
- **When inspecting the DLC-2, touch it with a paper clip or similar thin pin without directly inserting a tester into the terminals.**
- **Disconnect the negative battery cable before performing any work that requires handling of connectors.**

Step	Inspection		Action
1	INSPECT BETWEEN BCM AND INSTRUMENT CLUSTER FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the connector 2 and 3 which has BCM terminals 3C,3D and 2D,2C. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals F and E. • Is the voltage at DLC-2 terminals F and E the same?	Yes	Go to Step 17.
		No	Go to the next step.
2	INSPECT BCM FOR SHORT BETWEEN CIRCUITS • Inspect for continuity between BCM terminals 3C and 3D. • Is there continuity?	Yes	Replace the BCM because there is a short between circuits in the BCM. (See BODY CONTROL MODULE (BCM) REMOVAL/ INSTALLATION.)
		No	Go to the next step.
3	INSPECT BETWEEN CONNECTORS C-28 AND C-29 AND BCM FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Connect the connector 2 and 3 which has BCM terminals 3C,3D and 2D,2C. • Disconnect the connectors C-28 and C-29. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals F and E. • Is the voltage at DLC-2 terminals F and E the same?	Yes	Repair or replace the wiring harness between the connectors C-28 and C-29 and BCM because the wiring harness is shorted between circuits.
		No	Go to the next step.
4	INSPECT BETWEEN AUTO LEVELING CONTROL MODULE AND CONNECTORS C-28 AND C-29 FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Inspect for continuity between auto leveling control module terminals AB and AD. • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 6.

Step	Inspection		Action
5	INSPECT AUTO LEVELING CONTROL MODULE FOR SHORT BETWEEN CIRCUITS • Disconnect the auto leveling control module connector. • Inspect for continuity between auto leveling control module terminals AB and AD (wiring harness side). • Is there continuity?	Yes	Repair or replace the wiring harness between the auto leveling control module and connectors C-28 and C-29 because the wiring harness is shorted between circuits.
		No	Replace the auto leveling control module because there is a short between circuits in the auto leveling control module. (See AUTO LEVELING CONTROL MODULE REMOVAL/ INSTALLATION.)
6	INSPECT BETWEEN DC-DC CONVERTER (i-ELOOP) AND CONNECTORS C-28 AND C-29 FOR SHORT BETWEEN CIRCUITS • Inspect for continuity between DC-DC converter (i-ELOOP) terminals 1M and 1N. • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 8.
7	INSPECT DC-DC CONVERTER (i-ELOOP) FOR SHORT BETWEEN CIRCUITS • Disconnect the DC-DC converter (i-ELOOP) connector. • Inspect for continuity between DC-DC converter (i-ELOOP) terminals 1M and 1N (wiring harness side). • Is there continuity?	Yes	Repair or replace the wiring harness between the DC-DC converter (i-ELOOP) and connectors C-28 and C-29 because the wiring harness is shorted between circuits.
		No	Replace the DC-DC converter (i-ELOOP) because there is a short between circuits in the DC-DC converter (i-ELOOP). (See DC-DC CONVERTER (i-ELOOP) REMOVAL/ INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
8	INSPECT BETWEEN CONNECTOR C-01 AND CONNECTORS C-28 AND C-29 FOR SHORT BETWEEN CIRCUITS • Disconnect the connector C-01. • Connect the connectors C-28 and C-29. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals F and E. • Is the voltage at DLC-2 terminals F and E the same?	Yes	Repair or replace the wiring harness between the connector C-01 and connectors C-28 and C-29 because the wiring harness is shorted between circuits.
		No	Go to the next step.
9	INSPECT BETWEEN PCM AND CONNECTOR C-01 FOR SHORT BETWEEN CIRCUITS • Measure the voltage at PCM terminals 2C and 2D. • Is the voltage at PCM terminals 2C and 2D the same?	Yes	Go to Step 13.
		No	Go to the next step.
10	INSPECT BETWEEN CONNECTOR C-01 AND CONNECTORS C-31 AND C-32 FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Connect the connector C-01. • Disconnect the connectors C-31 and C-32. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals F and E. • Is the voltage at DLC-2 terminals F and E the same?	Yes	Repair or replace the wiring harness between the connector C-01 and connectors C-31 and C-32 because the wiring harness is shorted between circuits.
		No	Go to the next step.

Step	Inspection		Action
11	INSPECT BETWEEN TCM AND CONNECTORS C-31 AND C-32 FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Inspect for continuity between TCM terminals E and F. • Is there continuity?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between the connector C-01 and connectors C-31 and C-32 because the wiring harness is shorted between circuits.
12	INSPECT TCM FOR SHORT BETWEEN CIRCUITS • Disconnect the TCM connector. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Inspect for continuity between TCM terminals E and F (wiring harness side). • Is there continuity?	Yes	Repair or replace the wiring harness between the TCM and connectors C-31 and C-32 because the wiring harness is shorted between circuits.
		No	Replace the TCM because there is a short between circuits in the TCM. (See CONTROL VALVE BODY REMOVAL/INSTALLATION [CW6A-EL].) (See CONTROL VALVE BODY REMOVAL/INSTALLATION [EW6A-EL].)
13	INSPECT BETWEEN CONNECTOR C-30 AND CONNECTOR C-01 FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the connector C-30. • Connect the connector C-01. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals F and E. • Is the voltage at DLC-2 terminals F and E the same?	Yes	Repair or replace the wiring harness between the connector C-30 and connector C-01 because the wiring harness is shorted between circuits.
		No	Go to the next step.
14	INSPECT BETWEEN DSC HU/CM AND CONNECTOR C-30 FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Inspect for continuity between DSC HU/CM terminals AF and AC. • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 16.
15	INSPECT DSC HU/CM FOR SHORT BETWEEN CIRCUITS • Disconnect the DSC HU/CM connector. • Inspect for continuity between DSC HU/CM terminals AF and AC (wiring harness side). • Is there continuity?	Yes	Repair or replace the wiring harness between the DSC HU/CM and connector C-30 because the wiring harness is shorted between circuits.
		No	Replace the DSC HU/CM because there is a short between circuits in the DSC HU/CM. (See DSC HU/CM REMOVAL/INSTALLATION [R.H.D.].)
16	INSPECT PCM FOR SHORT BETWEEN CIRCUITS • Disconnect the PCM connector. • Inspect for continuity between PCM terminals 2C and 2D (wiring harness side). • Is there continuity?	Yes	Repair or replace the wiring harness between the PCM and connector C-30 because the wiring harness is shorted between circuits.
		No	Replace the PCM because there is a short between circuits in the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

Step	Inspection		Action
17	INSPECT BETWEEN CONNECTOR C-07 AND INSTRUMENT CLUSTER FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the connector C-07. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals F and E. • Is the voltage at DLC-2 terminals F and E the same?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between the BCM and connector C-07 because the wiring harness is shorted between circuits.
18	INSPECT BETWEEN CONNECTOR C-07 AND CONNECTOR C-11 FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Connect the connector C-07. • Disconnect the connector C-11. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at BCM terminals 2D and 2C. • Is the voltage BCM terminals 2D and 2C the same?	Yes	Repair or replace the wiring harness between the connector C-07 and connector C-11 because the wiring harness is shorted between circuits.
		No	Go to the next step.
19	INSPECT BETWEEN CONNECTOR C-11 AND INSTRUMENT CLUSTER FOR SHORT BETWEEN CIRCUITS • Measure the voltage at DLC-2 terminals F and E. • Is the voltage at DLC-2 terminals F and E the same?	Yes	Go to Step 26.
		No	Go to the next step.
20	INSPECT BETWEEN CONNECTORS C-51 AND C-52 AND CONNECTOR C-11 FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the connectors C-51 and C-52. • Connect the connector C-11. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals F and E. • Is the voltage at DLC-2 terminals F and E the same?	Yes	Repair or replace the wiring harness between the connectors C-51 and C-52 and connector C-11 because the wiring harness is shorted between circuits.
		No	Go to the next step.

Step	Inspection		Action
21	INSPECT BETWEEN LASER SENSOR AND CONNECTORS C-51 AND C-52 FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Inspect for continuity between laser sensor terminals D and C. • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 23.
22	INSPECT LASER SENSOR FOR SHORT BETWEEN CIRCUITS • Disconnect the laser sensor connector. • Inspect for continuity between laser sensor terminals D and C (wiring harness side). • Is there continuity?	Yes	Repair or replace the wiring harness between the laser sensor and connectors C-51 and C-52 because the wiring harness is shorted between circuits.
		No	Replace the laser sensor because there is a short between circuits in the laser sensor. (See LASER SENSOR REMOVAL/INSTALLATION.)
23	INSPECT BETWEEN FORWARD SENSING CAMERA (FSC) AND CONNECTORS C-51 AND C-52 FOR SHORT BETWEEN CIRCUITS • Inspect for continuity between forward sensing camera (FSC) terminals E and C. • Is there continuity?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between the connector C-11 and connectors C-51 and C-52 because the wiring harness is shorted between circuits.
24	INSPECT BETWEEN FORWARD SENSING CAMERA (FSC) AND CONNECTOR C-23 FOR SHORT BETWEEN CIRCUITS • Disconnect the connector C-23. • Inspect for continuity between forward sensing camera (FSC) terminals E and C. • Is there continuity?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between the connectors C-51 and C-52 and connector C-23 because the wiring harness is shorted between circuits.
25	INSPECT FORWARD SENSING CAMERA (FSC) FOR SHORT BETWEEN CIRCUITS • Disconnect the forward sensing camera (FSC) connector. • Inspect for continuity between forward sensing camera (FSC) terminals E and C (wiring harness side). • Is there continuity?	Yes	Repair or replace the wiring harness between the forward sensing camera (FSC) and connector C-23 because the wiring harness is shorted between circuits.
		No	Replace the forward sensing camera (FSC) because there is a short between circuits in the forward sensing camera (FSC). (See FORWARD SENSING CAMERA (FSC) REMOVAL/INSTALLATION.)
26	INSPECT BETWEEN CONNECTOR C-36 AND INSTRUMENT CLUSTER FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the connector C-36. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals F and E. • Is the voltage at DLC-2 terminals F and E the same?	Yes	Go to Step 33.
		No	Go to the next step.
27	INSPECT BETWEEN CONNECTIVITY MASTER UNIT (CMU) AND CONNECTOS C-36 FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Inspect for continuity between connectivity master unit (CMU) terminals 2E and 2F. • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 29.

Step	Inspection		Action
28	INSPECT CONNECTIVITY MASTER UNIT (CMU) FOR SHORT BETWEEN CIRCUITS • Disconnect the connectivity master unit (CMU) connector. • Inspect for continuity between connectivity master unit (CMU) terminals 2E and 2F (wiring harness side). • Is there continuity?	Yes	Repair or replace the wiring harness between the connectivity master unit (CMU) and connector C-36 because the wiring harness is shorted between circuits.
		No	Replace the connectivity master unit (CMU) because there is a short between circuits in the connectivity master unit. (See CONNECTIVITY MASTER UNIT (CMU) REMOVAL/ INSTALLATION.)
29	INSPECT BETWEEN EPS CONTROL MODULE AND CONNECTOR C-36 FOR SHORT BETWEEN CIRCUITS • Inspect for continuity between EPS control module terminals 1G and 1H. • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 31.
30	INSPECT EPS CONTROL MODULE FOR SHORT BETWEEN CIRCUITS • Disconnect the EPS control module connector. • Inspect for continuity between EPS control module terminals 1G and 1H (wiring harness side). • Is there continuity?	Yes	Repair or replace the wiring harness between the EPS control module and connector C-36 because the wiring harness is shorted between circuits.
		No	Replace the EPS control module because there is a short between circuits in the EPS control module. (See STEERING WHEEL AND COLUMN REMOVAL/ INSTALLATION.)
31	INSPECT BETWEEN CONNECTOR C-11 AND CONNECTOR C-36 FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Connect the connector C-11. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at BCM terminals 2D and 2C. • Is the voltage BCM terminals 2D and 2C the same?	Yes	Repair or replace the wiring harness between the connector C-11 and connector C-36 because the wiring harness is shorted between circuits.
		No	Go to the next step.
32	INSPECT SAS CONTORL MODULE FOR SHORT BETWEEN CIRCUITS • Disconnect the SAS control module connector. • Inspect for continuity between SAS control module terminals 3Q and 3S (wiring harness side). • Is there continuity?	Yes	Repair or replace the wiring harness between the SAS control module and connector C-36 because the wiring harness is shorted between circuits.
		No	Replace the SAS control module because there is a short between circuits in the SAS control module. (See SAS CONTROL MODULE REMOVAL/ INSTALLATION [TWO-STEP DEPLOYMENT CONTROL SYSTEM].) (See SAS CONTROL MODULE REMOVAL/ INSTALLATION [STANDARD DEPLOYMENT CONTROL SYSTEM].)
33	INSPECT BETWEEN CONNECTOR C-37 AND DLC-2 FOR SHORT BETWEEN CIRCUITS • Disconnect the connector C-37. • Inspect for continuity between DLC-2 terminals F and E. • Is there continuity?	Yes	Repair or replace the wiring harness between connector C-37 and DLC-2 because the wiring harness is shorted between circuits.
		No	Go to the next step.
34	INSPECT BETWEEN START STOP UNIT AND CONNECTOR C-37 FOR SHORT BETWEEN CIRCUITS • Is there continuity between start stop unit terminals 2M and 2O. • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 36.

Step	Inspection		Action
35	INSPECT START STOP UNIT FOR SHORT BETWEEN CIRCUITS • Disconnect the start stop unit connector. • Is there continuity between start stop unit terminals 2M and 2O (wiring harness side). • Is there continuity?	Yes	Repair or replace the wiring harness between the start stop unit and connector C-37 because the wiring harness is shorted between circuits.
		No	Replace the start stop unit because there is a short between circuits in the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.)
36	INSPECT BETWEEN CONNECTOR C-37 AND INSTRUMENT CLUSTER FOR SHORT BETWEEN CIRCUITS • Measure the voltage at instrument cluster terminals 1B and 1D. • Is the voltage instrument cluster terminals 1B and 1D the same?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between the connector C-36 and connector C-37 because the wiring harness is shorted between circuits.
37	INSPECT INSTRUMENT CLUSTER FOR SHORT BETWEEN CIRCUITS • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Connect the connector C-37. • Disconnect the instrument cluster connector. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals F and E. • Is the voltage at DLC-2 terminals F and E the same?	Yes	Repair or replace the wiring harness between the instrument cluster and connector C-37 because the wiring harness is shorted between circuits.
		No	Replace the instrument cluster because there is a short between circuits in the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)